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THOX Community Mesh: Unlinkable Subscriber Tokens, Digest-Audited Stage Execution and a Witness-Cosigned Settlement Ledger for Member-Contributed Inference

How MeshStack™ subscribers can pool their own hardware, and reach cloud models through it, without the pool learning who is asking: a design, a working virtual lab and the measurements behind both.

Craig Ross and Tommy Xaypanya · Corresponding author: Tommy Xaypanya, CTO and Co-Founder, THOX.ai LLC · ORCID 0009-0000-4425-9565

ABSTRACT

Most people who want to run a capable language model can afford neither the hardware nor a metered cloud account, while their neighbours' computers sit idle. This paper specifies and tests a community mesh for THOX MeshStack™: members add their own devices to a shared pool, and any subscriber can run distributed inference on that pool, or reach a hosted cloud model through it, without the pool learning who is asking. The design combines blind-signed subscriber tokens (RFC 9474) and sealed envelopes through a relay (RFC 9180) for requester unlinkability; signed per-hop receipts with sampled replay audits for integrity; contribution credits that raise a member's token budget and tier, with no currency; and a witness-cosigned Merkle ledger that settles credits and penalties after the request, off the token path. We state the limit plainly: a contributor that executes a stage can read its activations, so the pool gives identity privacy, not content privacy, and requests that need the latter are refused rather than routed. On a real 135M-parameter model split into three stages, a token-exact re-execution audit, the natural choice, exposed 69 of 102 cheated requests and missed an fp16 weight substitution on all 17 prompts; a SHA-256 digest of the stage boundary exposed 102 of 102. Minting an unlinkable token costs 7.9 ms at RSA-2048 and verifying one 0.12 ms; the ledger costs 0.92 microseconds per receipt. A virtual lab of 15 processes with real cryptography ran 12 scenarios, including a contributor killed mid-request, an owner reclaiming a device, a cheating contributor caught and penalised, and a forged ledger head refused by every witness; all passed and every lab request was token-identical to the whole model. A simulation shows the cost of unlinkability for fairness: fixed issuance budgets isolate heavy users but waste capacity, and indexing issuance to pool load recovers the waste. This edition is consolidated with a companion THOX paper and lab on the same subject: we adopt its plan-wide admission rule, execution contexts and replica-first default, reproduce its 62 tests and 150-trial campaign on a second host, and measure the replica-first claim in our own lab (6.6% lower per-remote-call round trip for one contributor holding both remote ranges than for two contributors in sequence on loopback, 30 matched pairs; corrected in v1.1.1, see Section 6.7). Nothing here is a THOX device, wide-area or throughput result.

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